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Depth based Hand Gesture Recognition for Smart Teaching

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***Abstract*—Gesture recognition plays a very important role in human-computer interaction, and depth based gesture recognition receives more attention because depth sensors have the advantages of capturing depth information and being robust to illumination changes. At present, gesture recognition algorithms focus on the accuracy and efficiency of recognition on general data sets, but ignore the specific needs of interactive gestures in specific scenarios, and the general gesture data sets can not meet the actual interactive needs, which also limits the application and promotion of human-computer interaction. Aiming at the above problems, this paper creates a specific hand gesture data set, which dedicated to interactive teaching of intelligent classroom teaching, and proposes a deep neural network model which integrates global and local information for gesture recognition. The experimental results demonstrate that the proposed deep model achieves 93.6% recognition rate of 17 commonly used gestures and verifies the performance in virtual geometry teaching.**

1. Introduction

Gesture recognition is of great importance to human-computer interaction. Gesture-based interaction technology makes up more than 50% of human-computer interaction technology [1]. In vision-based gesture recognition, its performance is influenced by illumination changes, occlusion and complex background. Depth sensors have the advantages of robust for the change of illumination, depth based gesture recognition has received more attention recently.

At present, most hand gesture recognition algorithms have done a lot of work for higher accuracy and efficiency. Relevant research work has achieved better recognition performance on commonly used gesture data sets [2]. Although the existing algorithms have achieved satisfactory experimental results on these data sets, the gesture based human-computer interaction technology has not been widely applied. One of the main reasons is that different human-computer interaction scenarios have special requirements for gestures, and the existing data sets cannot meet this demand. Moreover, most of the algorithms based on deep learning do not take into account the fusion of global and local information of human hands, which has a negative impact on the performance of gesture recognition.

Intelligent classroom is a potentially important scenario of gesture interaction [3], [4]. Gesture provides a natural interaction channel in teaching, which could replace the mouse and keyboard to interact with virtual objects and PPT, so that teachers can focus on teaching course and improve the effect of classroom teaching. In this paper, novelty gesture data set is created for intelligent classroom teaching, and a deep neural network model integrating global and local information is proposed. The model is validated in the virtual geometry teaching.

The organization of this paper is as follows, section 2 introduces the related work of gesture recognition, the novel gesture data set and the proposed deep neural network are presented in section 3. In section 4, experimental results are demonstrated, and conclusions are given in section 5.

2. Related Work

As a natural and intuitive channel of communication, hand gesture is very popular in human-computer interaction. The immersive virtual system based on gesture recognition could effectively complete the tasks of target selection, roaming and virtual object manipulation. Gesture recognition has attracted many researchers.

With the success of deep learning in image recognition and wide use of depth sensors, researchers have proposed many gesture recognition methods based on depth information and deep neural networks.

Koller et al present a video based hand shape classifier, which was based on weakly labelled sequence data by embedding a CNN within an iterative EM algorithm [5]. The proposed classifier was evaluated on two publicly available large sign language data sets for continuous sign language recognition. Sinha et al estimate the 3D hand pose using depth data, and train convolutional neural networks to obtain spatio-temporal activation features. Their method achieved real time performance compares to state-of-the-art methods [6]. Molchanov et al proposed a drivers hand gesture recognition method from challenging depth and intensity data using 3D convolutional neural networks, which achieved a correct classification rate of 77.5% on the VIVA challenge dataset [7]. In Ref [8], authors proposed a DLEH2 feature for depth-based dynamic hand gesture recognition. DLEH2 feature fused hand shape and spatio-temporal features of

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depth sequences. Experimental results on two public benchmark datasets show that their fusion scheme outperforms the state-of-the-art methods on public data sets. Yang et al used a recurrent 3D convolutional neural network for dynamic hand gesture recognition from multi-modal data. The authors also introduced a multimodal hand gesture dataset captured with depth, color and stereo-IR sensors, and their method achieved an accuracy of 83.8% on that data set [9]. Hsiao et al present a Large RGB-D hand gesture data set, LaRED, which has 81 classes and each gesture data is composed of color images, depth images and masks of hand region [10]. The paper also proposed a baseline recognition algorithms based on LaRED.

Although many gesture recognition algorithms and challenging data sets have been proposed, the application and promotion of gesture interaction technology still suffer from the following problems: (1) General data sets can not satisfy the needs of specific interaction scenarios for gesture types; (2) Depth-based algorithms seldom consider the fusion of global and local information of gesture, which affects the accuracy of gesture recognition.

3. The proposed method

3.1. Hand gesture data set

Hand gesture based interaction plays an important role in intelligent teaching. Gesture could replace mouse and keyboard to generate interactive commands, so that teachers can use gesture to control PPT or interact with virtual objects. This paper creates a hand gesture data set containing 17 kinds of static gestures according to the actual teaching needs.

Hand gesture data is captured via Kinect 2.0. More than 700 people participated in the collection work. The database used in this experiment consists of 510,000 depth image data, and each gesture includes 30,000 samples. 17 gestures represent interactive commands such as selection, seizure, release, number and execution. The segmentation of hand gesture is shown in Figure 1, and samples are shown in Figure 2.

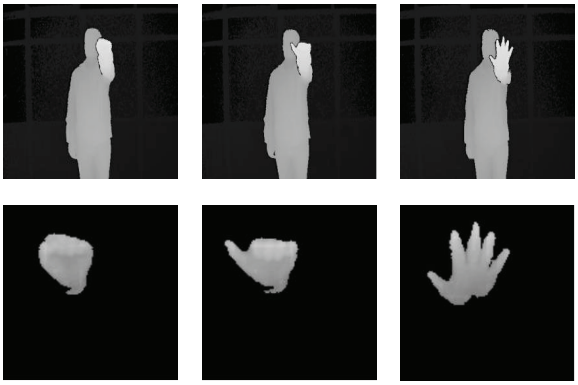

Figure 1. The segmentation of hand gesture.

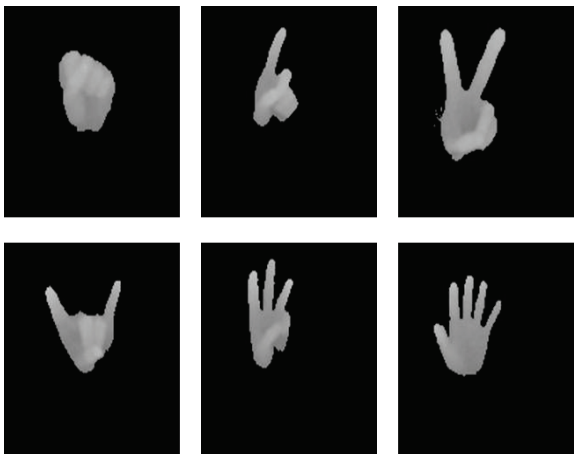

Figure 2. Hand gesture samples.

3.2. Hand gesture recognition

The RGB-D data captured by Kinect are pre-processed and segmented to generate the depth gesture data. The deep neural network fuses the data of the final convolution layer and the previous pooling layer, which produces the full connection layer features based on the global and local features of gesture. The gesture recognition results will be obtained via Softmax, which has the loss function as Equation 1.

$$L(W) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} log(P(Y=k|X^{(n)}))^{g(Y^{(n)},k)} + \frac{\lambda}{2}\sum_{i=1}^{K}\sum_{j=0}^{M} w_{ij}^2 \quad (1)$$

where,

$$g(Y^{(n)},k) = f(x) = \{^{1,Y^{(n)}=k}_{0,Y^{(n)}\neq k} \quad (2)$$

In order to prevent over-fitting, the weight attenuation term is added to regulate the parameter W. In our experiment, the weight attenuation term is 0.0005. In this paper, the batch stochastic gradient descent algorithm is used to optimize the loss function, and the back-propagation algorithm is used to iteratively update the parameters.

Different from other deep based gesture recognition methods, the proposed deep neural network integrates global and local information of hand gesture. The data set used for training is large and the network is more complex. Figure 3 shows the flowchart of gesture recognition. Firstly, the RGB-D data is captured by Kinect, then the segmented gesture images are extracted and used as input of the deep neural network (network structure is shown in Table 1). The trained deep neural network classifies the hand gestures into different categories.

As shown in Table 1, the deep neural networks has 8 layers, which includes 5 convolution layers and 3 full connection layers. The inputs images are normalized to

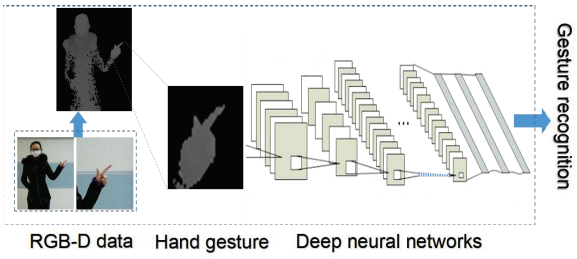

Figure 3. The flowchart of hand gesture recognition.

200×200 three-channel images, which are the inputs of Conv1 layer, and Conv1 has 96 trainable convolutional kernels, the kernel size is 11×11. The sliding window convolution operation is then performed on the input image with a step size of 4, and 96 feature maps are obtained correspondingly, size of each feature map is 48×48. The feature maps will be corrected via ReLU activation function, that is, the negative response value is corrected to 0 and the positive response value remains unchanged. Then, the output of ReLU will be normalized via a LRN layer. The pooling layer is utilized to reduce the dimension of feature maps. Later convolution layers (Conv2, Conv3, Conv4 and conv5) are similar to Conv1 layer, which have different parameters. The size of convolutional kernel for Conv2 is 5×5, and the step size is 1. The ReLU function, LRN layer and pooling layer, which are used in Conv1 will be also applied to Conv2. In particular, LRN layer is removed in conv5 layer, and only ReLU layer is added behind Conv3 and Conv4. Thus, the proposed network combines the advantages of local and global gesture information, and is more effective for feature learning. In order to reduce the risk of overfitting, a dropout layer is added between the full connection layer FC6 and FC7.

LRN(Local Response Normalization) is helpful to improve the generalization of the model. $a^i_{x,y}$ denotes the activated value of i convolution kernel at (x, y). The value of local response normalization, $b^i_{x,y}$, is achieved by Equation 3.

$$b^i_{x,y} = a^i_{x,y}/(k + a\sum\nolimits_j^m (a^i_{x,y})^2)^\beta \tag{3}$$

where N is the total number of feature maps, $j = max(0, i-n/2)$ and $m = min(N-1, i+n/2)$. Parameters in Equation 3 are set as follows, $k = 2, n = 5, a = 10^{-4}, \beta = 0.75$.

4. Experimental results

In experiments, the input image is pre-processed to 200×200 size, and batch stochastic gradient descent algorithm is used to optimize the loss function (Equation 1), and back-propagation algorithm is used to tune the network parameters. The Bach-size is set to 128, and the initial learning rate is 0.001. The dropout rate is set to 0.5 in the dropout layer and the weight parameters are all initialized to 0.01, and the bias is all initialized to 0. Our model was trained on Tesla K40 GPU server. Three kinds of commonly used neural networks are used for comparative experiments. Table 2 shows the parameter settings of the networks and the accuracy of gesture recognition.

TABLE 1. THE STRUCTURE OF THE PROPOSED NEURAL NETWORK

Layer	Input	Kernel_size	Stride	Output
Conv1	3@200x200	11x11	4	96@48x48
Pool1	96@48x48	3x3	2	96@24x24
Conv2	96@24x24	5x5	1	256@24x24
Pool2	3@256@24x24	3x3	2	256@12x12
Conv3	256@12x12	3x3	1	384@12x12
Conv4	384@12x12	3x3	1	384@12x12
Conv5	384@12x12	3x3	1	256@12x12
Pool5	256@12x12	3x3	2	256@6x6
FC6	9216			512
FC7	512			512
FC8	512			6

TABLE 2. COMPARISONS OF HAND GESTURE RECOGNITION

	base_lr	max_iter	batch_size	Accuracy
CaffeNet	0.001	120000	300	90.49%
AlexNet	0.001	190000	300	91.14%
GoogleNet	0.001	80000	100	91.87%
Proposed	0.001	100000	128	93.60%

When CaffeNet reaches 76500 iterations, the accuracy of gesture recognition reaches 90.49%. Other networks are similar to it. Addressing the accuracy of gesture recognition, the three networks are similar to each other. The proposed method achieves higher accuracy because it integrates the global gesture information. If the size of training data set is increased, the proposed method can achieve better performance. The training time costing of the neural network is greatly influenced by the network structure. The training process of CaffeNet takes about 37 hours. Although the other networks reduce the number of iterations, the training time does not decrease.

According to Figure 4, loss value on validation set stops updating after 30,000 iterations, and the model was converged. For better generalization, we choose the model with 13,000 iterations. The loss value of the 13,000 iterations model on the validation set is 0.0691878, and the correct rate is up to 98.0556%.

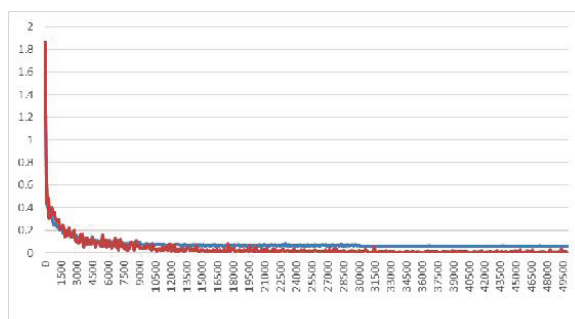

Figure 4. The loss value of validation set(Blue) and training set(Red).

The recognition rates of different types of gestures are also quite different. For example, the fourteenth gesture is like the number 7, which causes occlusion during rotation and making the gesture very similar to the first $fist$ gesture. The recognition rate of fourteenth gesture is only 89%. The first $fist$ gesture and the six gesture are not easy to cause self-occlusion, and the recognition rate of these gesture is as high as 98%.

The complexity and occlusion of gesture shape have great influence on the accuracy of gesture recognition. Figure 5 shows the recall and precision recognition rates of hand gestures in Figure 2. The gestures, i.e.,$g1$, $g4$ and $g6$ have higher precisions, because the number of these gestures that misclassified into other categories is relatively small. $g3$ and $g4$ have lower recall rates, which is due to the fact that $g2$ maybe misclassified into the categories of $g3$ and $g4$, and $g3$ maybe misclassified into the categories of $g4$ and $g5$.

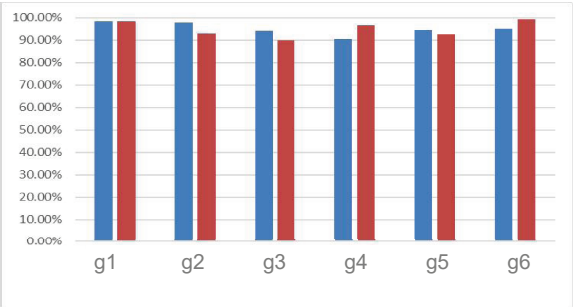

Figure 5. The recall and precision rates of hand gestures recognition, $g1$ through $g6$ denote the gestures showed in Figure 2.

In addition, low quality of hand segmentation and self-occlusion could lead to a smaller discrimination between gesture categories, which will reduce the accuracy of gesture recognition. Figure 6 shows some gestures with unsatisfactory segmentation, and gestures with self-occlusion. As Figure 6 demonstrated, $g1$ has a higher recognition accuracy, because no self-occlusion occurred in $fist$. The first image in second row demonstrates a $g2$ gesture occluded by itself, which will be classified into $g1$ category.

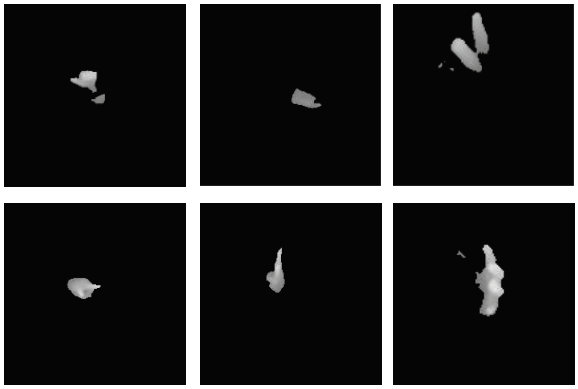

Figure 6. Hand gestures with unsatisfactory segmentation are shows in the first row, and second row shows hand gestures with self-occlusion.

The proposed method was also integrated into a virtual geometry teaching framework, which achieved real-time performance of human-computer interaction.

5. Conclusions

In this paper, a depth gesture data set for interactive teaching scenarios is proposed, and a deep neural network that integrates global and local features of hand gesture is designed for gesture recognition. The experimental results show that the proposed method achieves higher accuracy and real-time performance when the proposed method was applied to the virtual geometry teaching.

Acknowledgments

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